PROJECT

Database Security System with Advanced Protection Mechanisms

Second Semester Academic Year 2526

Member 1: ______________________________ Course/Year/Section: ______________________
Member 2: ______________________________ Date: ______________________
Member 3: ______________________________
Member 4: ______________________________
Member 5: ______________________________

Project Overview

The goal of this project is to design and develop a secure database system that implements advanced security mechanisms to protect sensitive data. The system focuses on safeguarding user credentials and critical information stored in the database through encryption, secure authentication, and access control techniques.

This project will incorporate database security principles, including data encryption, secure authentication methods such as Single Sign-On (SSO), One-Time Password (OTP), and AppKey validation. It also ensures that sensitive data is protected both in transit and at rest, following industry-standard security practices.

Objectives

The key objectives of this project are:

- Design a secure database system with authentication mechanisms such as OAuth, SSO, AppKey, and OTP.
- Ensure strong database security through encryption of stored data and secure access control.
- Protect user credentials using advanced hashing algorithms such as bcrypt or Argon2.
- Implement OTP verification as an additional layer of database access security.
- Apply secure database configurations, including SSL/TLS encryption for connections.
- Provide complete documentation explaining database security implementation and best practices.

Project Deliverables

The deliverables for this project include:

1. Secure Database System
    a. A database that securely stores user credentials, AppKeys, and sensitive data.
    b. Proper database schema with access control and normalization.

2. Authentication System
    a. Login system integrated with SSO providers (Google, Apple, Facebook, etc.).
    b. AppKey validation for secure application access.
    c. OTP verification for multi-factor authentication.

3. Data Encryption
    a. Implementation of password hashing using bcrypt or Argon2.
    b. Encryption of sensitive data stored in the database.

4. Database Security
    a. Secure database connections using SSL/TLS.
    b. Role-based access control (RBAC).

5. Project Document
    a. Step-by-step setup guide.
    b. Explanation of database security techniques used.
    c. Reflection on the importance of database security.

6. Testing and Debugging
    a. Security testing
    b. Functional testing of authentication and database operations.

7. Final Report
    a. Summary, challenges encountered, and solutions.
    b. Importance of database security in real-world systems.

Project Scope

Languages and Tools:

- Programming Languages: Python, PHP, JavaScript, Node.js
- Database: MySQL, PostgreSQL, or MongoDB
- Libraries/Frameworks: bcrypt, Argon2, OAuth APIs, Twilio/SendGrid for OTP
- Hosting: AWS, Firebase, Heroku, or Local Server

Core Features

- **Data Encryption:** Sensitive data is encrypted before storage.
- **Password Security:** Passwords are hashed and salted using secure algorithms.
- **Secure Authentication:** Multi-layer authentication (SSO + OTP + AppKey).
- **Database Protection:** Prevention of SQL Injection and unauthorized access.
- **Secure Communication:** Use of SSL/TLS for database connections.

Team Structure

The project will be completed by a group of five (5) members:
Member 1 (Frontend Developer)
• Design the user interface for login and authentication pages.
• Implement input validation and password complexity rules.
• Integrate SSO authentication (Google, Facebook, etc.).

Member 2 (Backend Developer)
• Develop authentication logic and server-side validation.
• Implement OTP functionality using APIs.
• Handle secure communication between application and database.

Member 3 (Database Administrator)
• Design and manage the secure database structure.
• Implement encryption for stored data.
• Configure secure database access and SSL/TLS connections.

Member 4 (Security Specialist)
• Identify vulnerabilities and perform security audits.
• Implement protection against SQL Injection and attacks.
• Ensure compliance with database security best practices.

Member 5 (Documentation & Testing)
• Document system setup and security implementation.
• Perform system testing and debugging.
• Ensure all requirements are met.

Timeline and Milestones
Week 1: Project planning, database design, and frontend setup
Week 2: Backend development and authentication implementation
Week 3: Database security and encryption
Week 4: Testing, debugging, documentation, and presentation

Budget and Resources
- Software: Open-source tools (Node.js, MySQL, etc.)
- Hosting: AWS Free Tier, Firebase, or Heroku
- APIs: Twilio/SendGrid (may include free-tier usage)

Evaluation Criteria
- **Functionality and Security:** System effectively secures database and authentication processes
- **Encryption and Data Protection:** Proper implementation of encryption and hashing techniques
- **Documentation Quality:** Clear and complete documentation of the system

Criteria	Description	Points
Functionality & Features	Secure database system with authentication, OTP, and AppKey fully implemented	35
Security & Encryption	Strong database security practices including encryption and attack prevention	40
Documentation & Team Output	Complete documentation and active participation of all members	25